Md. Abdullah Al Mahmud

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Backend-focused Software Engineer with 7+ years of experience in architecting resilient, distributed systems and high-throughput microservices. Proven track record in scaling infrastructure and optimizing data pipelines for real-world production environments.

EXPERIENCE

Independent Software Engineer & Technical Consultant — *Self-Employed*

JANUARY 2023 - PRESENT

- **Systems & Tooling:** Engineered cross-platform CLI developer utilities in Rust and high-throughput backend services in Python and Go, implementing AI-native workflows to accelerate R&D.
- **Architecture & Infrastructure:** Configured containerized microservices and benchmarked database query performance, maintaining local Linux/DevOps testbeds for high-availability database and queue architectures.

Neahtid — *Software Engineer*

OCTOBER 2021 - SEPTEMBER 2022

- Architected high-performance RESTful APIs and relational schemas for multi-tenant applications; mentored junior engineers, enforced code review standards, and set CI/CD practices to elevate team velocity.

VivaSoft Limited — *Software Engineer*

JANUARY 2020 - FEBRUARY 2021

- Engineered microservices for a large-scale educational platform (Campus.com), developing reusable internal libraries and CLI tools that streamlined development workflows.
- Led a cross-functional team in developing an automated IoT-driven vertical gardening system (Grönska), architecting a custom high-performance REST framework that optimized API response times by ~20% for real-time sensor data telemetry.

Jobike Limited — *Software Engineer*

JUNE 2018 - MARCH 2019

- Architected real-time fleet tracking infrastructure, queue-based SMS authentication pipelines for scale onboarding, and idempotent mobile payment integration microservices (bKash).

GoBD — *Software Engineer*

JULY 2015 - JANUARY 2017

- Engineered real-time transit tracking microservices, custom Raspberry Pi vehicle tracking hardware, and automated systemd server-health monitoring tools.

EDUCATION

American International University-Bangladesh

BSc. in Computer Science and Engineering - *(APRIL 2011 - JANUARY 2016)*

PROJECTS

- **GraphRAG Exploration:** Developed a PoC for semantic retrieval using Neo4j and LangGraph, exploring integration patterns for knowledge-graph-based RAG.
- **Logistics Heuristic Engine:** Architected a multi-stop routing optimization engine, designing high-performance heuristics for real-world spatial constraints.

SKILLS

Languages: Python (Expert), Rust, Go, JavaScript, TypeScript

Backend & Data: Django (Expert), FastAPI, PostgreSQL, Redis, RESTful API Design, Schema Design, ETL

Infrastructure: AWS (EC2, ECS, Auto-scaling), Docker, Linux, CI/CD, Monitoring & Alerting

Engineering Core: Microservices, System Architecture, Performance Optimization, Distributed Systems

Additional Exposure: Java (Spring), MongoDB, Elasticsearch, GraphQL

ACTIVITIES

Participated in the
ACM ICPC Dhaka Regionals

- In **2013,** with the team ***Algorithm_The_Podokkhep***
- and in **2014** with team ***MangoBar***.
- Participated in **NCPC 2014** with team ***_fatalError***

INTERESTS

- Music / Playing Guitar
- Travelling / Hiking
- Electronics projects
- Painting and Sketching

ONLINE PRESENCE

- LinkedIn
- GitHub
- StackOverflow
- Personal Blog
- LeetCode
- HackerRank